

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 10 - Unit Test 3

Unit Test 3 - Answer Key

Class: Class 10

Assessment Type: Unit Test 3

Total Marks: 10

Duration: 30 minutes

Topics Covered: Python Programming

ICT Teaching Resources Portal

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Section I: MCQ (5 x 1 = 5 marks)

1. **Answer:** c - YTH

- **Explanation:** `s[1:4]` extracts characters from index 1 to 3 (excluding 4). "PYTHON" → index 1='Y', 2='T', 3='H' → "YTH".

2. **Answer:** a - {}

- **Explanation:** {} creates an empty dictionary. [] creates a list, () creates a tuple.

3. **Answer:** b - Removes leading and trailing whitespace

- **Explanation:** `strip()` removes whitespace from both ends of a string. It does not remove spaces in the middle.

4. **Answer:** b - re

- **Explanation:** Python's built-in module for regular expressions is `re`. It provides functions like `match()`, `search()`, `findall()`.

5. **Answer:** a - [1, 2, 3, 1, 2, 3]

- **Explanation:** `[1, 2, 3] * 2` repeats the list twice, resulting in `[1, 2, 3, 1, 2, 3]`.

Section II: Short Answer (5 x 1 = 5 marks)

1. **Output and explanation**

- **Answer:**

1. apple
2. banana
3. cherry

`enumerate()` returns index-item pairs. `i` gets the index (0,1,2), `fruit` gets the value. `i+1` starts numbering from 1. f-string formats the output.

- **Marking:** 1 mark for output, 1 mark for explanation.

2. **split() vs join()**

- **Answer:** `split()` divides a string into a list based on a delimiter. Example: "hello world".`split()` → ["hello", "world"]. `join()` combines list elements into a string with a separator. Example: " ".`join(["hello", "world"])` → "hello world".
- **Marking:** 1 mark for each method with example.

3. **Palindrome function**

- **Answer:**

```
def is_palindrome(s):  
    s = s.lower()  
    return s == s[::-1]
```

```
# Test  
print(is_palindrome("Madam")) # True
```

```
print(is_palindrome("hello")) # False
print(is_palindrome("Racecar")) # True
```

- **Marking:** 1 mark for function, 0.5 marks for each test case.

4. List comprehension vs Generator expression

- **Answer:** List comprehension creates a complete list in memory: `[x**2 for x in range(10)]`. Generator expression yields items lazily: `(x**2 for x in range(10))`. Generators are memory-efficient for large data. List comprehensions are faster for small datasets.
- **Marking:** 1 mark for difference, 1 mark for examples.

5. Dictionary output

- **Answer:**

```
['a', 'b', 'c']
[1, 2, 3]
0
```

`d.keys()` returns all keys, `d.values()` returns all values, `d.get("d", 0)` returns 0 (default) since "d" is not in the dictionary.

- **Marking:** 0.5 marks for each correct output.